

DRISHTI

Intelligent Conversational AI & Crime Analytics Platform

Karnataka State Police — AI Components Technical Guide

DRISHTI is a crime-intelligence platform that combines 8 specialized AI systems into one unified interface. Each AI component solves a distinct real-world police challenge — from reading a scanned FIR in Kannada, to mapping the hidden network behind a drug cartel, to predicting next week's crime hotspot. This document explains all 8 AIs with plain-language descriptions and concrete Karnataka Police use-case examples.

Table of Contents

- [AI #1 RAG / Conversational LLM](#) — Chat-based FIR queries
- [AI #2 OCR](#) — Optical Character Recog. — Read scanned Kannada documents
- [AI #3 NER](#) — Named Entity Recognition — Extract people, places, dates
- [AI #4 Time Series Forecasting](#) — Predict crime spikes
- [AI #5 Anomaly Detection](#) — Catch unusual overnight surges
- [AI #6 Document / Identity AI](#) — Aadhaar, passbook, cheque scan
- [AI #7 Case Similarity / MO Matching](#) — Link crimes by pattern
- [AI #8 Graph / Network Analysis AI](#) — Map criminal networks

RAG — Retrieval-Augmented Generation (Conversational LLM)

Tool: Catalyst QuickML RAG + LLM Serving

■ What It Does

Lets police officers type plain-language questions — in English or Kannada — and get structured, source-backed answers drawn directly from police databases. No SQL knowledge required. The system finds relevant records, reads them, and writes a human-readable response with full citations.

■ How It Works

The officer's question is converted into a vector (numerical representation). The system searches the database for the most relevant FIRs, case files, and accused records. Those records are passed to an LLM which then composes a clear answer, always citing the source record numbers.

■ Karnataka Police Use-Case Example

Scenario: Inspector Verma at Banaswadi PS wants a quick history of a suspect named Raju Kumar before conducting an interrogation.

Officer Input / System Trigger:

Officer types: "Show me all FIRs where Raju Kumar was accused in last 2 years, Bangalore."

System Output:

DRISHTI Response: Found 4 FIRs matching your query: FIR #1023 — Whitefield PS — Robbery — Jan 2024 — Chargesheeted FIR #2341 — Banaswadi PS — Theft — Aug 2023 — Under Investigation FIR #0892 — Shivajinagar — Assault — Mar 2023 — Convicted FIR #3012 — Mysuru North — Dacoity — Nov 2022 — Absconding Sources: CCTNS records verified (4 documents) [Export as PDF] [Ask a follow-up question]

■ Why It Matters for Police

Without this AI, an inspector would need to call 4 different police stations, wait for callbacks, and manually compile the history — taking hours. DRISHTI delivers the same result in under 30 seconds, fully sourced.

Real-World Analogy: Just like asking ChatGPT a question and getting a sourced answer — but here it queries actual live Karnataka Police FIR databases instead of the internet.

OCR — Optical Character Recognition

Tool: Catalyst Zia OCR (supports Kannada + Indian languages)

■ What It Does

Reads scanned FIR documents, handwritten complaint letters, and photo evidence — including text written in Kannada script — and converts them into searchable, structured digital text that the rest of the system can analyse.

■ How It Works

An image or scanned PDF is uploaded to the system. Zia OCR analyses pixel patterns to recognise characters in both English and Kannada. The extracted text is then cleaned, normalised, and fed into the NER and search pipeline for further analysis.

■ Karnataka Police Use-Case Example

Scenario: A 2019 FIR from Gulbarga PS only exists as a paper scan. A new case may be linked to it, but the old record is not searchable in CCTNS.

Officer Input / System Trigger:

Officer uploads: scanned_FIR_Gulbarga_2019.pdf System trigger: OCR pipeline auto-starts on upload

System Output:

OCR Extraction Complete: Language detected: Kannada + English (mixed) Pages processed: 3
Confidence: 94.2% Extracted fields: FIR Number : 0234/2019 Date Filed : 14-Mar-2019
Accused Name: ■■■■ ■■■■■■ (Raju Kumar) Crime Type : Theft under IPC 379 Location :
Near Bus Stand, Gulbarga Record now indexed and searchable in DRISHTI.

■ Why It Matters for Police

Thousands of pre-digital FIRs across Karnataka exist only on paper. OCR makes them instantly searchable, allowing investigators to discover connections to old cases that would otherwise remain invisible.

Real-World Analogy: Like Google Lens reading a restaurant menu in Japanese — but here it reads a handwritten Kannada FIR and makes it searchable in a database.

NER — Named Entity Recognition

Tool: Catalyst Zia Text Analytics

■ What It Does

Automatically reads unstructured case narrative text and extracts structured entities: people's names, locations, dates, phone numbers, vehicle numbers, organisation names, and amounts — without any manual data entry.

■ How It Works

After OCR converts a document to text, NER scans every sentence to classify words into entity types. It understands context — so 'Raju' in one sentence is marked as a person, while 'Banaswadi' is marked as a location, and '14 January 2024' as a date, all automatically.

■ Karnataka Police Use-Case Example

Scenario: A complaint narrative has been typed or OCR-extracted. It reads: 'On 14th January at 8 AM, Raju Kumar and his associate Shiva met near Koramangala flyover. They used a white Maruti Suzuki KA-05-MN-4421 and contacted via 9876543210.'

Officer Input / System Trigger:

Input text (from OCR or typed FIR narrative – auto-processed by pipeline)

System Output:

NER Extraction Result: PERSON : Raju Kumar, Shiva LOCATION : Koramangala flyover, Bangalore DATE/TIME : 14th January, 8 AM VEHICLE : KA-05-MN-4421 (Maruti Suzuki, White) PHONE : 9876543210 Auto-linked: Raju Kumar → matches accused in FIR #1023, #2341 KA-05-MN-4421 → flagged in FIR #3012 (Mysuru, 2022) 9876543210 → registered to: Ali Baig

■ Why It Matters for Police

Manual data entry from FIR narratives is error-prone and time-consuming. NER does it in milliseconds and immediately cross-links the extracted entities against the entire existing database — surfacing connections a human might miss.

Real-World Analogy: Like how Gmail automatically detects a flight booking in an email and adds it to Google Calendar — but here it reads a police complaint and automatically tags every person, place, and vehicle mentioned.

Time Series Forecasting — Crime Trend Prediction

Tool: Catalyst QuickML Time Series Models

■ What It Does

Analyses historical crime data over time to identify seasonal patterns, weekly cycles, and long-term trends — then forecasts where and when crime is likely to rise in the coming days or weeks.

■ How It Works

Years of FIR data are aggregated by station, beat, crime type, day, and hour. The forecasting model learns recurring patterns — like how thefts spike every Sunday morning, or how chain snatching surges during festival seasons. It then projects these patterns forward to generate advisory risk forecasts.

■ Karnataka Police Use-Case Example

Scenario: The DCP of Bangalore East wants to plan patrol deployment for the upcoming Dasara festival week without waiting for crimes to actually happen.

Officer Input / System Trigger:

Supervisor Dashboard – Forecast View Selected: Bangalore East District | Next 7 days | All crime types

System Output:

7-Day Crime Forecast – Bangalore East: Chain Snatching : HIGH RISK (▲ 210% above baseline) Peak window : 6 PM – 9 PM, MG Road & Brigade Road Historical basis: Same pattern in Dasara 2022, 2023, 2024 Vehicle Theft : MODERATE RISK (▲ 80%) Peak window : 11 PM – 2 AM, parking areas near venues Pickpocketing : HIGH RISK near Cubbon Park & Lalbagh Recommended action: Deploy 6 additional motorcycle units on MG Road corridor from 5 PM–10 PM, Oct 12–16. Confidence: 87%

■ Why It Matters for Police

Police today deploy patrol based on intuition and past experience. This AI makes deployment data-driven — reducing crime by putting officers where the data says they're needed, before the crime happens.

Real-World Analogy: Like how weather apps predict rain 7 days in advance using historical patterns — this AI predicts crime 7 days in advance using historical FIR data.

Anomaly Detection — Unusual Surge Alert System

Tool: Catalyst QuickML Anomaly Detection

■ What It Does

Continuously monitors incoming FIR data in near-real-time and raises an automatic alert the moment any station or area shows a crime surge significantly above its historical normal — even at 3 AM when no supervisor is actively watching.

■ How It Works

The system maintains a rolling baseline for each police station: average FIRs per hour, per crime type, per day-of-week. Every hour it compares current incoming FIRs against this baseline. If the deviation exceeds a threshold (e.g. 3x normal), it fires an alert with full evidence attached.

■ Karnataka Police Use-Case Example

Scenario: Between 11 PM and 2 AM on a Tuesday, Hebbal PS registers 9 vehicle theft FIRs. The normal average for that time slot is 2. No supervisor noticed because it was overnight.

Officer Input / System Trigger:

System auto-trigger – Nightly anomaly scan job (runs every hour via scheduler)

System Output:

ALERT GENERATED – 02:14 AM, 31 May 2026 Station : Hebbal PS, Bangalore North Crime type : Vehicle Theft FIRs filed : 9 in last 3 hours Normal avg : 2 per night (same time window) Deviation : +350% above baseline Pattern match: Similar MO to Yelahanka cluster (Nov 2023 – 11 vehicles stolen in 4 nights) Suggested action: Check NH-44 service road CCTV. Evidence: FIR #5601, #5602, #5603 ... (+6 more) AI Confidence: 91% [Acknowledge] [Assign Investigating Officer] [View All FIRs]

■ Why It Matters for Police

A gang can steal 5 vehicles in one night and be gone before morning briefing. This AI catches the pattern within hours — giving police a chance to respond the same night instead of discovering it in the morning report.

Real-World Analogy: Like your bank's fraud detection system — it automatically flags when your card is used in an unusual location at 3 AM and blocks it immediately, without waiting for you to notice.

Document AI — Identity & Financial Document Scanner

Tool: Catalyst Zia Identity Scanner (IN data centre)

■ What It Does

Extracts structured data from Indian identity documents and financial records: Aadhaar cards, PAN cards, bank passbooks, and cheque leaves. Converts a photographed document into structured fields that can be directly linked to accused records, FIRs, and transaction networks.

■ How It Works

A photograph or scan of the document is passed to the Identity Scanner. Specialised models trained on Indian document formats extract fields like name, Aadhaar number, account number, IFSC code, and transaction amounts. The output is a structured JSON object that is automatically linked to the relevant person and FIR in the database.

■ Karnataka Police Use-Case Example

Scenario: During a hawala investigation, an officer seizes a bank passbook belonging to a suspect. The passbook shows large irregular cash deposits followed by immediate withdrawals — a possible money laundering pattern.

Officer Input / System Trigger:

Officer uploads photo: suspect_passbook_canara_bank.jpg

System Output:

Document AI — Passbook Extraction Result: Bank : Canara Bank, Rajajinagar Branch Account Name : Suresh Kumar Account No. : XXXX XXXX 4521 IFSC : CNRB0001234 Flagged Transactions: 14 Jan 2024 — CREDIT ₹85,000 — Sender: Unidentified 03 Feb 2024 — CREDIT ₹1,20,000 — Sender: Unidentified 05 Feb 2024 — DEBIT ₹2,05,000 — Cash withdrawal AI Pattern Flag: Large cash withdrawals within 48hrs of deposits. Possible hawala pattern detected. Cross-reference: Matches transaction pattern in FIR #2341. [Link to FIR #2341] [Add to Case File]

■ Why It Matters for Police

Forensic financial analysis that previously required a specialist accountant and several days of work can now be initiated in under 2 minutes — directly from a photograph taken on the officer's phone at the scene.

Real-World Analogy: Like how a bank's KYC system instantly reads your Aadhaar and fills in all your details when you open an account — but here it reads a seized document and connects it to a criminal investigation.

Case Similarity AI — Modus Operandi (MO) Matching

Tool: LLM Semantic Embeddings + Structured Feature Matching

■ What It Does

Compares a current FIR against thousands of past cases to find crimes committed using the same method, time pattern, or behavioural signature — even across different police stations or districts — to identify serial offenders and link related cases.

■ How It Works

Each case is represented as a combination of structured features (crime type, time, location, entry method, victim profile) and a semantic vector from the narrative text. When a new FIR is filed, the system computes similarity scores against all existing cases and returns the top matches ranked by confidence.

■ Karnataka Police Use-Case Example

Scenario: Three house break-ins are registered at different Bangalore stations in one week. No investigator has connected them because they're at different police stations and assigned to different officers.

Officer Input / System Trigger:

Chat query: "Find cases similar to FIR #5701 — house break-in, Indiranagar"

System Output:

MO Similarity Results for FIR #5701: Match 1: FIR #5698 — Domlur PS — 3 days ago
Similarity: 91% Common factors: Sunday morning, ground floor, bathroom window entry, no violence, electronics stolen Match 2: FIR #5645 — Marathahalli PS — 8 days ago
Similarity: 88% Common factors: Same time window, same entry method, same victim profile (working couple, 8 AM-10 AM) Match 3: FIR #4921 — Whitefield PS — 3 months ago (unsolved)
Similarity: 79% Recommendation: High probability of serial offender. Suggest joint investigation across 3 stations. [Create Linked Case Group] [Assign Lead Investigator]

■ Why It Matters for Police

Serial criminals operate across jurisdictions precisely because police station boundaries create blind spots. This AI eliminates those blind spots by treating the entire city's FIR database as one searchable intelligence asset.

Real-World Analogy: Like how Spotify's recommendation engine finds songs 'similar' to what you played — but here it finds crimes 'similar' in pattern to a new FIR, helping detectives link a serial burglar's work across the city.

Graph AI — Criminal Network Analysis

Tool: Graph Algorithms: Degree Centrality, Community Detection (Python on AppSail)

■ What It Does

Builds a visual network graph connecting every accused person, victim, phone number, vehicle, address, bank account, and FIR — and then uses graph algorithms to surface hidden relationships, identify key brokers in criminal networks, and discover organised crime clusters.

■ How It Works

All entities (persons, devices, locations, accounts) are nodes. All relationships (co-accused, shared phone, shared address, same vehicle) are edges. Graph algorithms then calculate centrality (who is most connected), betweenness (who bridges two otherwise separate groups), and community detection (which people form a cluster or gang).

■ Karnataka Police Use-Case Example

Scenario: A drug trafficking suspect named Shiva is arrested in Bangalore. Investigators want to know his full network — who else is involved, what other cases are connected, and who the key organiser is.

Officer Input / System Trigger:

Chat query: "Show criminal network for accused: Shiva, FIR #4521"

System Output:

Network Graph Generated — Shiva's Network: Nodes identified: 7 persons, 3 phones, 2 vehicles, 1 bank account, 5 FIRs Direct connections from Shiva: Ramesh Kumar — co-accused (FIR #4521, #3012) Ali Baig — shared phone 98765XXXX KA-09-AB-1234 — vehicle used in FIR #4521 & #2876 Key finding — Bridge node detected: Ali Baig appears in 4 separate unrelated FIRs across 3 different police stations. Centrality score: CRITICAL Role: Likely network coordinator / broker Community cluster: 3 persons form a tight sub-group consistent with a drug distribution cell. Suggested action: Prioritise surveillance on Ali Baig. Request CDR (Call Detail Records) for 98765XXXX. [View Full Graph] [Export Network Report] [Flag for surveillance]

■ Why It Matters for Police

Organised crime groups deliberately use intermediaries to avoid direct connections. This AI finds those intermediaries automatically — the 'Ali Baig' types who appear in multiple cases but have never been the primary accused. Without graph analysis, these connectors remain invisible.

Real-World Analogy: Like how LinkedIn shows 'mutual connections' and '2nd degree connections' — but here it maps criminal connections across FIRs, exposing the real organiser behind a network of accused who appear to be unrelated.

#	AI Component	Catalyst Tool	Kaam (Function)
1	RAG / LLM	QuickML RAG	Plain-language FIR queries with cited answers
2	OCR	Zia OCR	Read scanned & Kannada documents
3	NER	Zia Text Analytics	Extract persons, places, dates, vehicles from text
4	Time Series Forecasting	QuickML Time Series	Predict crime spikes before they happen
5	Anomaly Detection	QuickML Anomaly	Alert on unusual overnight FIR surges
6	Document AI	Zia Identity Scanner	Extract data from Aadhaar, passbooks, cheques
7	MO Matching	LLM Embeddings + ML	Link serial crimes by pattern across stations
8	Graph AI	Graph Algorithms (Python / AppSail)	Map hidden criminal networks & key brokers

All 8 AI components work together inside DRISHTI. Remove any one and the platform loses a critical investigative capability. Together they transform Karnataka Police from reactive record-keeping to proactive, AI-powered crime intelligence.